

Unit 2
Science stage 6
Topic 2.1
Properties of
substances

Lesson Objectives

By the end of the lesson, I will know

6Cp.01 Know that the temperature at which a substance changes state is a property of the substance.

6Cc.03 Describe the difference between boiling and evaporation.

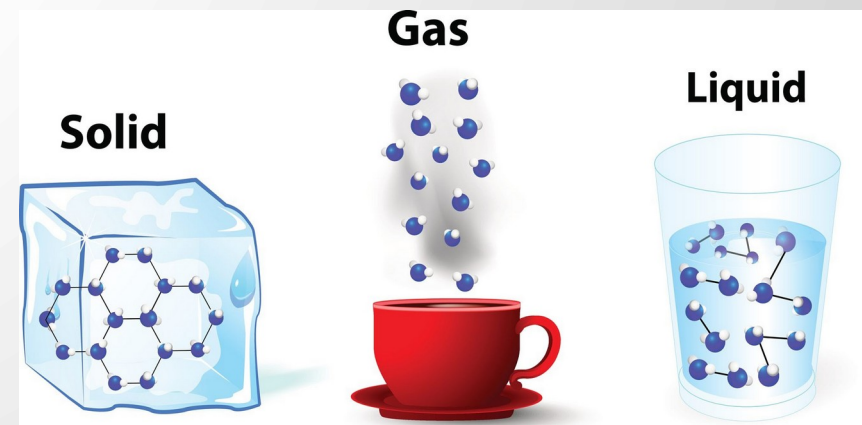
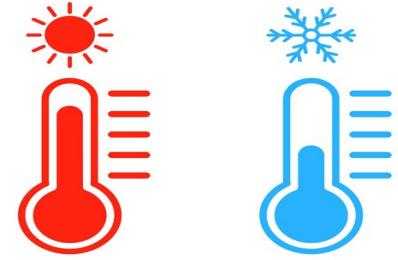
6Cp.02 Know that gases have properties, including mass.

Success Criteria

- I can understand that the temperature at which a substance changes state is a property of the substance.

- I can describe the difference between boiling and evaporation.

- I can identify properties of gases.



New Vocabulary-Notebook

1. Melting points
2. Boiling point
3. Properties
4. Evaporation
5. Mass

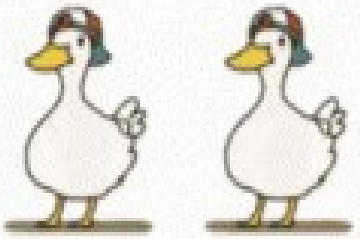
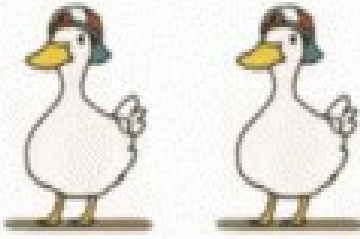
Please note- add definitions to the new terms using words you can understand

01 Warm up

What is a Gas?

What is a Liquid?

What is a solid?

SOLID MOLECULES	LIQUID MOLECULES	GAS MOLECULES
		
		

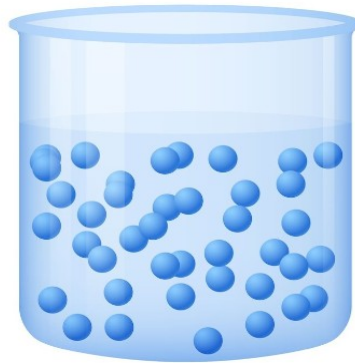
State of substances

SOLID



- Rigid
- Fixed Shape
- Fixed Volume
- Cannot be squashed

LIQUID



- Not Rigid
- No Fixed Shape
- Fixed Volume
- Cannot be squashed

GAS



- Not Rigid
- No Fixed Shape
- No Fixed Volume
- Can be squashed

What do you notice?

Explain what is happening to the water?



Introductory video

**WHAT IS
FREEZING,
MELTING AND
BOILING POINT?**



Myth Buster!

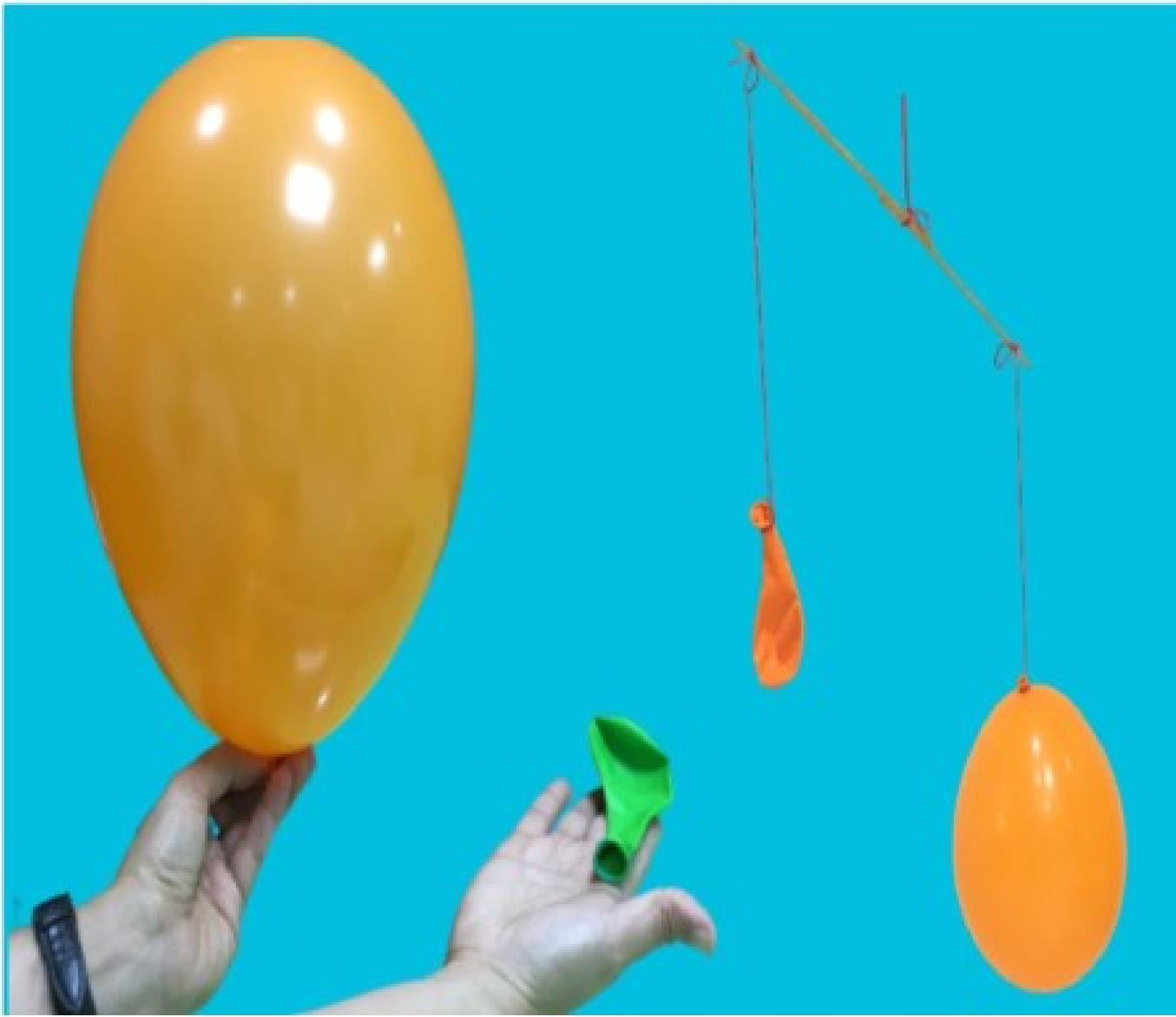


many people make mistakes here!

Do gasses have mass?



to help us to understand it more effectively.



If gas didn't have mass, the blown up balloon would have an equal weight to the empty balloon.

Mass

We process information to help us to understand it more effectively.

Melting point

Water has a melting point of 0°C which means

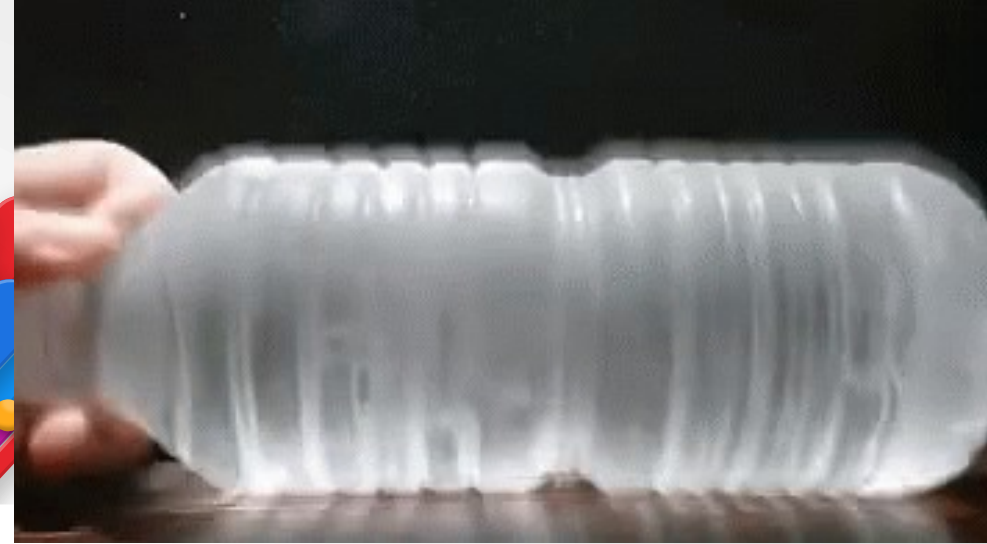
At 0°C water will turn into a liquid when it was frozen

The temperature at which something becomes liquid



Freezing point

The temperature at which something becomes solid



— Irene M. Pepperberg

Evaporation

is a substance's change of phase from liquid to gas.



liquid

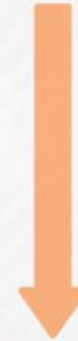


gas

Evaporation is a type of vaporization.



heat

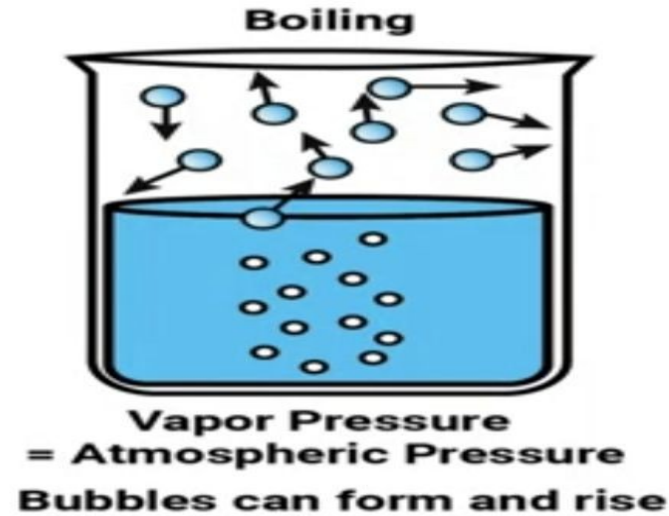
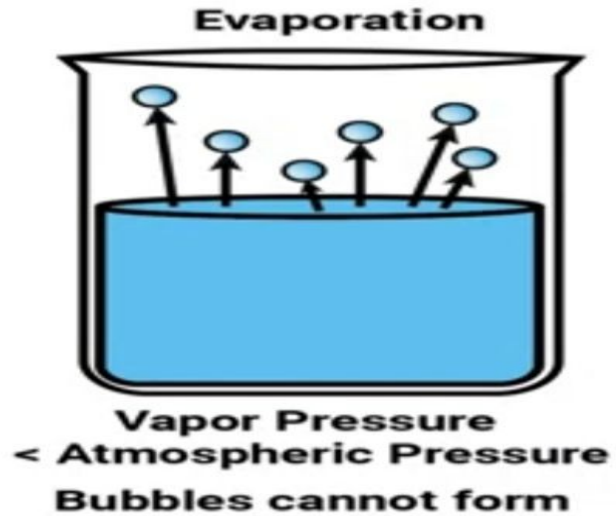


gas

evaporation



Evaporation Vs Boiling

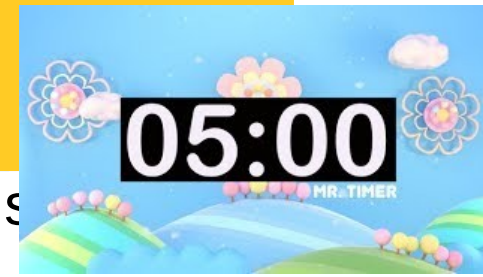


Evaporation	Boiling
No bubbles. ⁺ ☺	Bubbles form throughout liquid.
Takes place only on surface of liquid.	Takes place throughout the liquid.
Slower process.	Faster process.
Occurs at any temperature.	Boils a specific temp.(boiling point).
Temperature can fluctuate .	Temperature remains constant .
Thermal energy from surroundings .	Thermal energy from external source .

Similarities:

- Both involve liquids changing to gases.
- Both need energy to take place.
- Both are physical changes.

QUICK TRUE OR FALSE

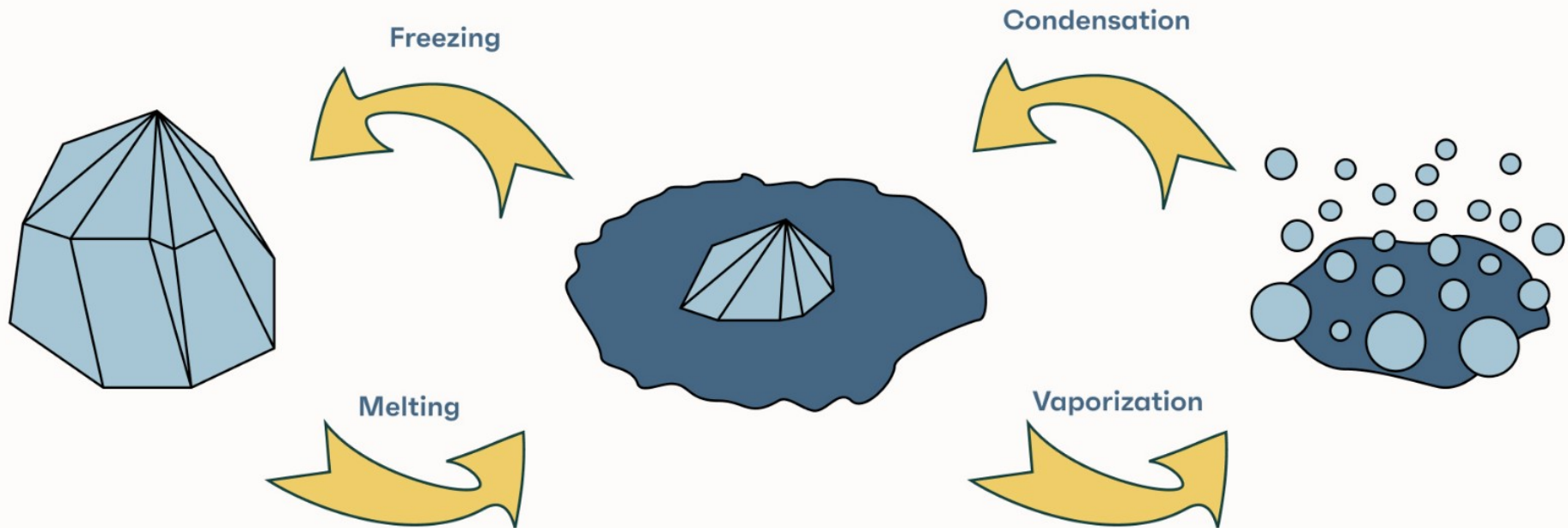


1. The temperature at which a substance changes state is a property of that substance. (False/True)
2. Evaporation only happens when a liquid reaches its boiling point. (False/True)
3. Boiling occurs at a fixed temperature for a given substance. (False/True)
4. Gases do not have mass because we cannot see them. (False/True)
5. Both boiling and evaporation change a liquid into a gas. (False/True)
6. The boiling point of water is always the same under normal conditions. (False/True)
7. Evaporation can happen at any temperature below the boiling point. (False/True)
8. When a balloon is inflated, its mass stays exactly the same as when it was empty. (False/True)

Solid

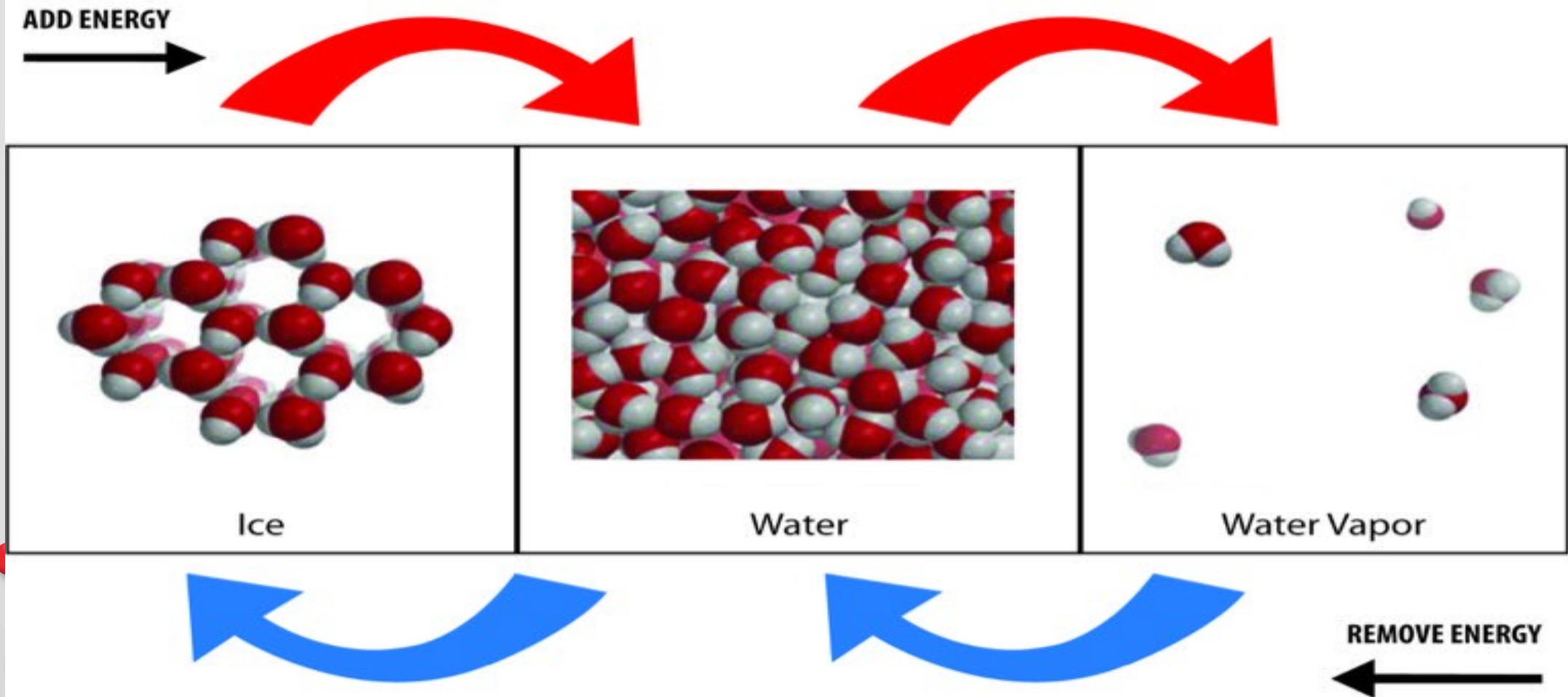
Liquid

Gas



Changes of State- Notebook Activity

Write down the name of the changes in the red arrows and the blue arrows



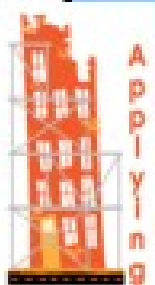


Data for experiment

Substances in the experiment

Substance	Melting Point (°C)	Observations (e.g., fast, slow, smoky)
Ice	0°C	Melted very quickly into water.
Butter	34°C	Melted into a yellow, oily liquid.
Beeswax	63°C	Took the longest time to melt.

Comparing Melting Points



Think like a scientist 1

Measure and compare melting points

You will need: three different solids, three pans, a hotplate, a thermometer, a stopwatch or digital watch

This activity is a teacher demonstration.

Safety: Do not touch the hot plate or pan: it can burn you.

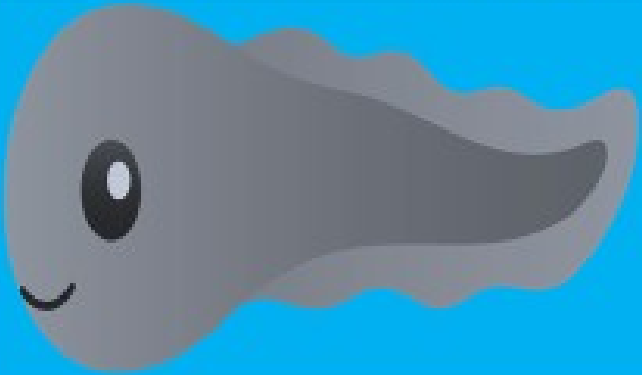
You teacher will do the following:

- 1 Place a solid in a pan.
- 2 Heat the solid substance until it melts. Remove the pan from the hot plate as soon as the substance melts.
- 3 Measure the temperature of the melted substance. You should record the reading in a table.
- 4 Repeat steps 1–3 with the two other solids.



We apply our learning to demonstrate understanding and then reflect on this.

Differentiation Levels!



Tadpole Level

- 1 Which substance had:
 - a the highest melting point?
 - b the lowest melting point?
- 2 Draw a bar chart of the results.



Frog Level

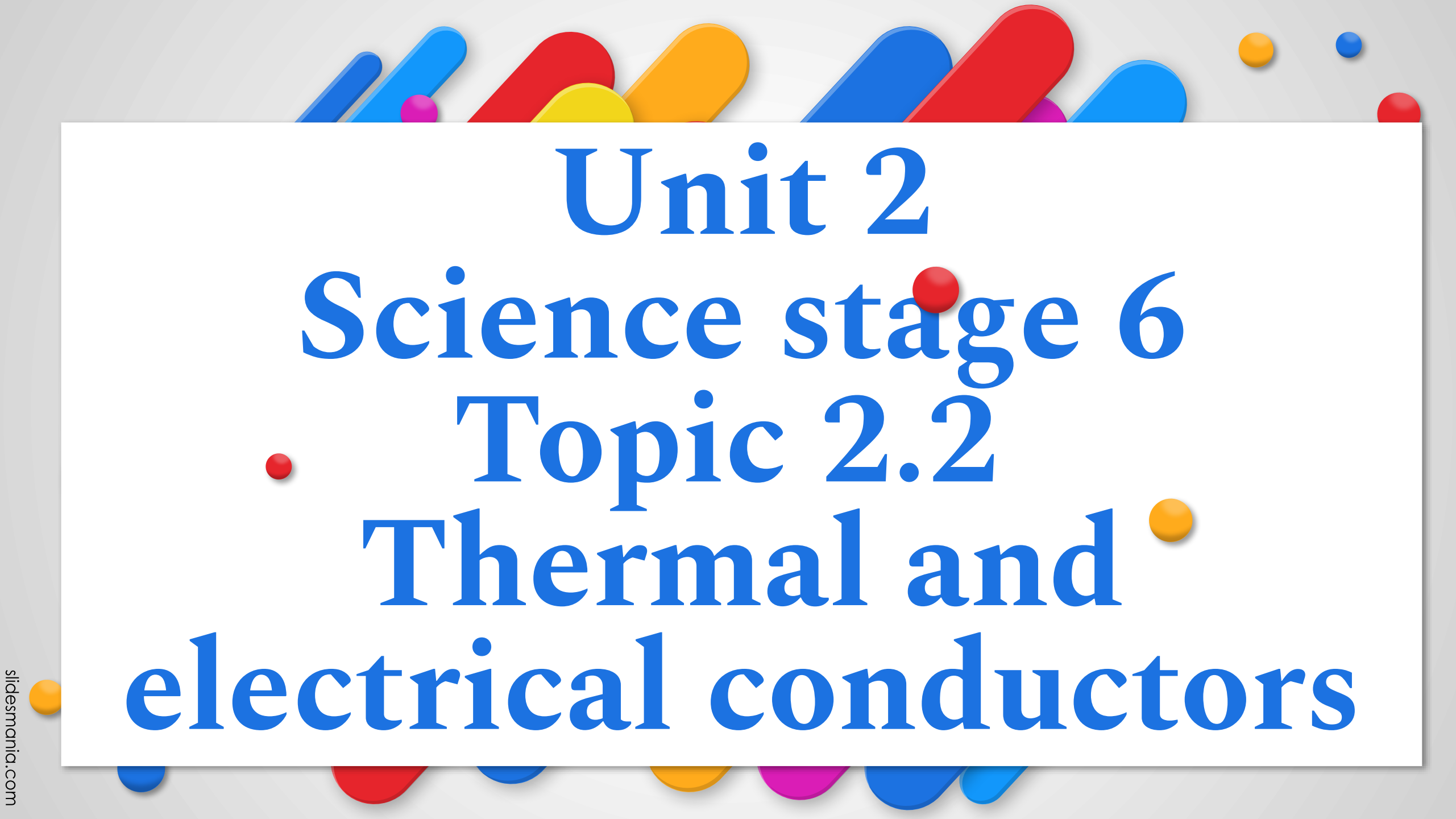
- 3 Write a conclusion for the investigation.
Remember a conclusion is what you found out in the investigation.
- 4 Would the melting points change if you heated more of the substance in each pan? Explain your answer.
- 5 Why is the melting point of a substance always lower than its boiling point?



Let's complete a definition match

- Blood
- Conduction
- Pulse
- Properties
- Circulation
- Thermal Conductor
- Boiling Point
- Circulatory System
- Electrical Conductor
- Melting Point

- The temperature at which a liquid becomes a gas.
- The temperature at which a solid becomes a liquid.
- Something about the substance that can be used to describe it.
- The movement of heat or electricity through matter.
- Materials and substances which conduct heat well.
- Materials and substances which conduct electricity well.
- The pumping of blood around the body.
- A small beat that can be felt due to the pressure of blood as the heart pumps blood around the body.
- The system formed by the heart, blood vessels and blood to carry food and oxygen around the body.
- A red liquid that carries food and oxygen to all parts of the body.



Unit 2

Science stage 6

Topic 2.2

Thermal and electrical conductors

Lesson Objectives

By the end of the lesson, I will know

6Cp.01 Know that the temperature at which a substance changes state is a property of the substance.

6Cc.03 Describe the difference between boiling and evaporation.

6Cp.02 Know that gases have properties, including mass.

New Vocabulary

1. Thermal conductor
2. Electrical conductor
3. Mass

What are material properties?

Hard



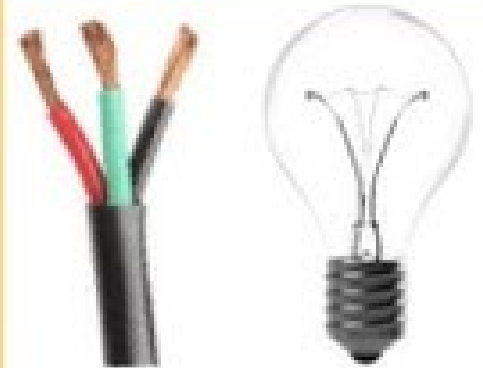
Soluble



Transparent



Conductor



Soft



Insoluble



Opaque



Insulators



Pronounce and understand today's keywords

Properties



Shiny

Hard

Transparent



Soft



Opaque

Dull

We gather all of the information that we need in order to learn.

Physical Properties

- Can be observed or measured without changing the matter's identity.

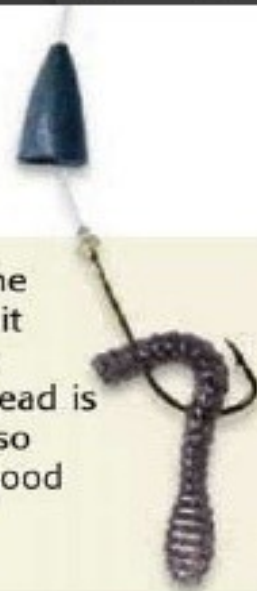
Examples of Physical Properties



Thermal conductivity (KAHN duhk TIV uh tee) is the rate at which a substance transfers heat. Plastic foam is a poor conductor.



State is the physical form in which a substance exists, such as a solid, liquid, or gas. Ice is water in the solid state.



Density is the mass per unit volume of a substance. Lead is very dense, so it makes a good sinker for a fishing line.



Solubility (SAHL yoo BIL uh tee) is the ability of a substance to dissolve in another substance. Flavored drink mix dissolves in water.

Ductility (duhk TIL uh tee) is the ability of a substance to be pulled into a wire. Copper is often used to make wiring because it is ductile.



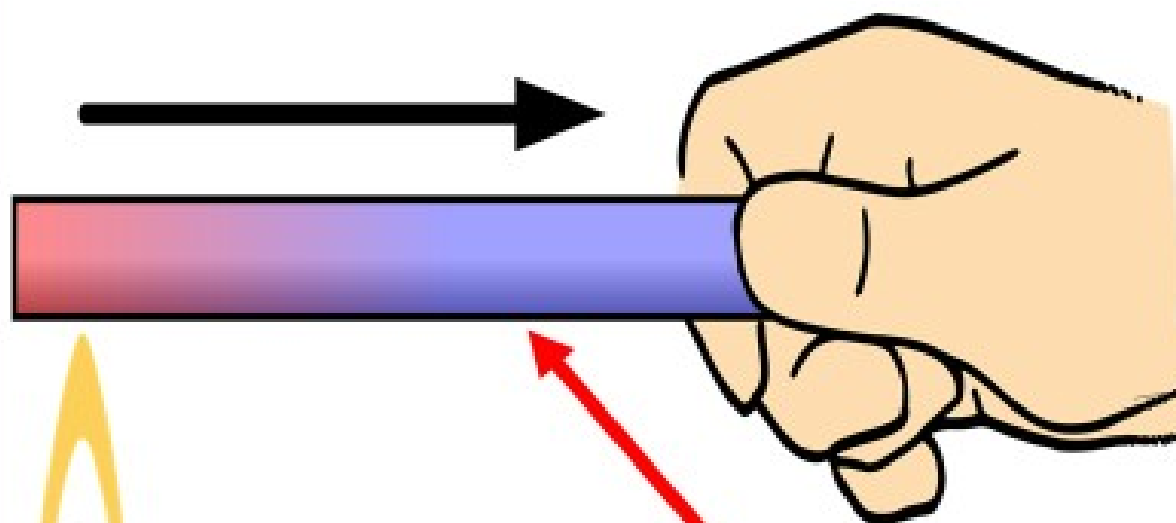
Malleability (MAL ee uh BIL uh tee) is the ability of a substance to be rolled or pounded into thin sheets. Aluminum can be rolled into sheets to make foil.



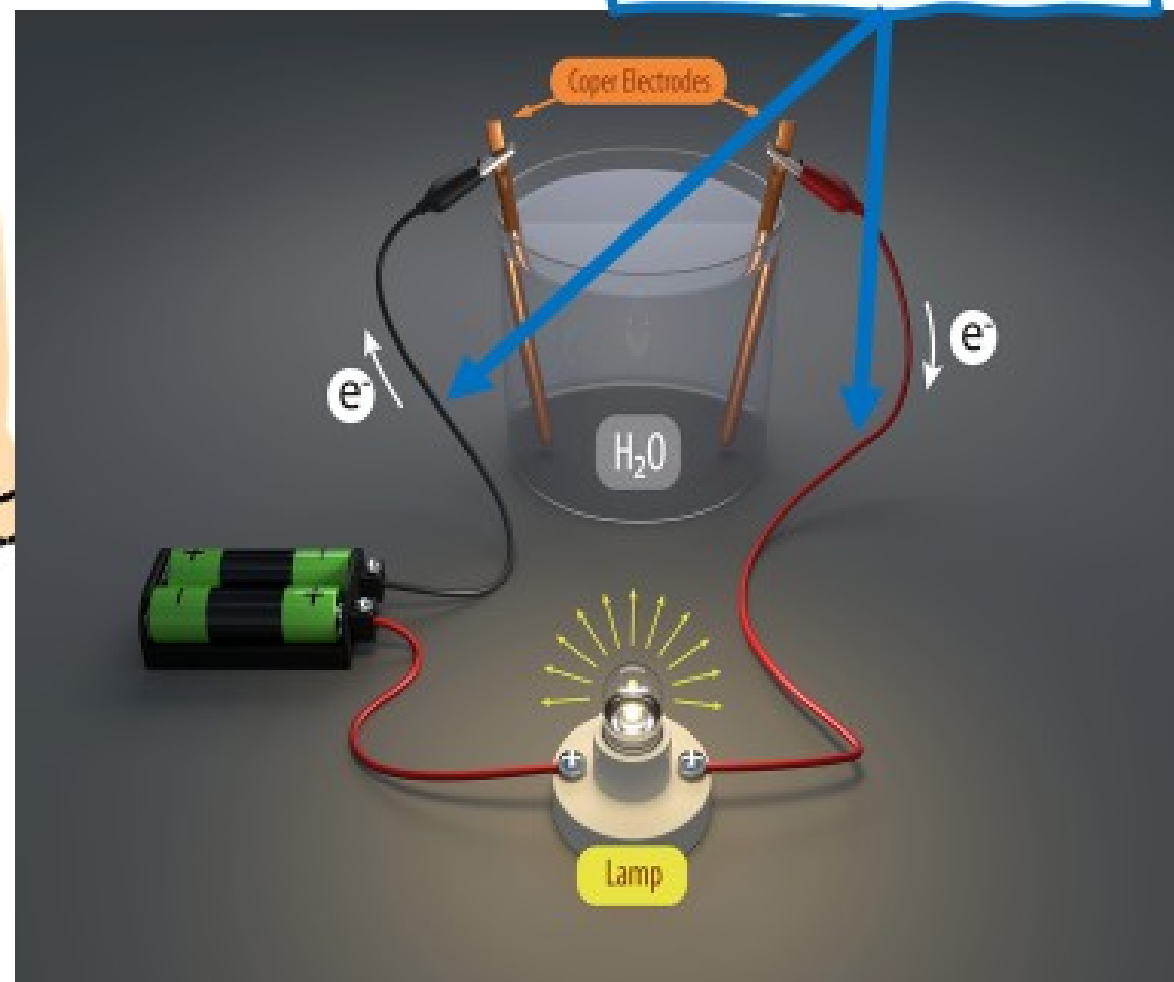
Pronounce and understand today's keywords

Conduction

Electrical
Conductor



Thermal
Conductor



We gather all of the information that we need in order to learn.